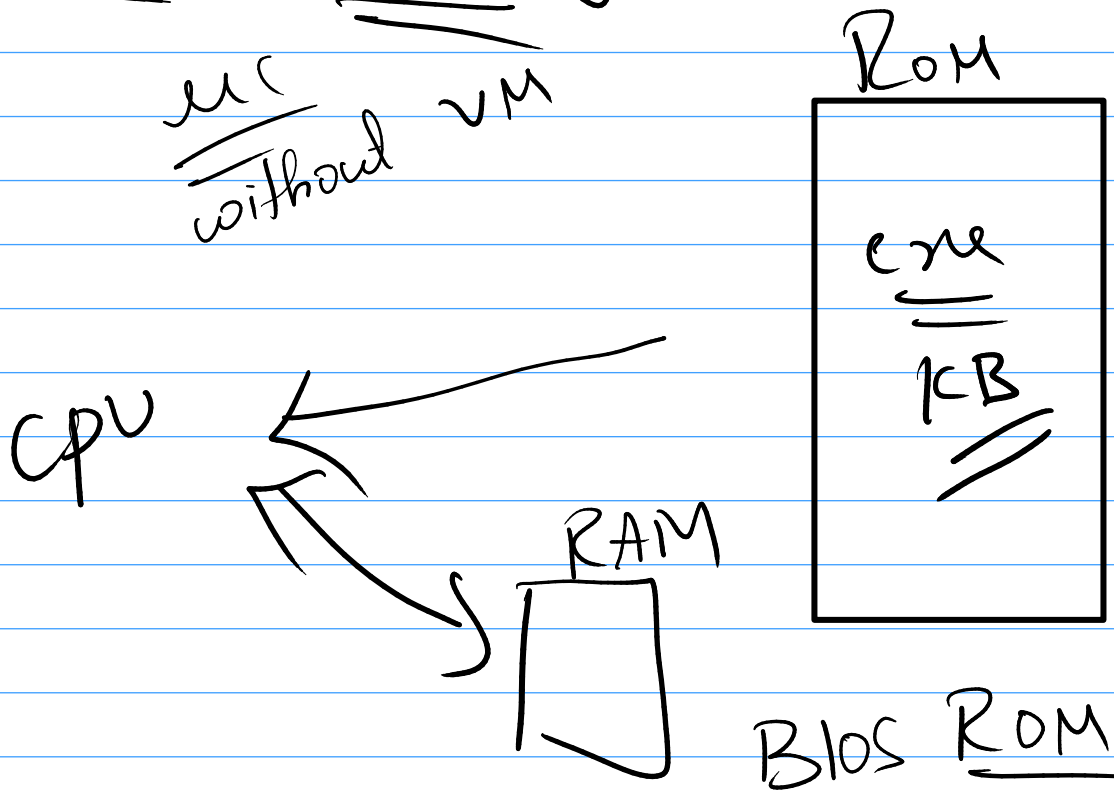
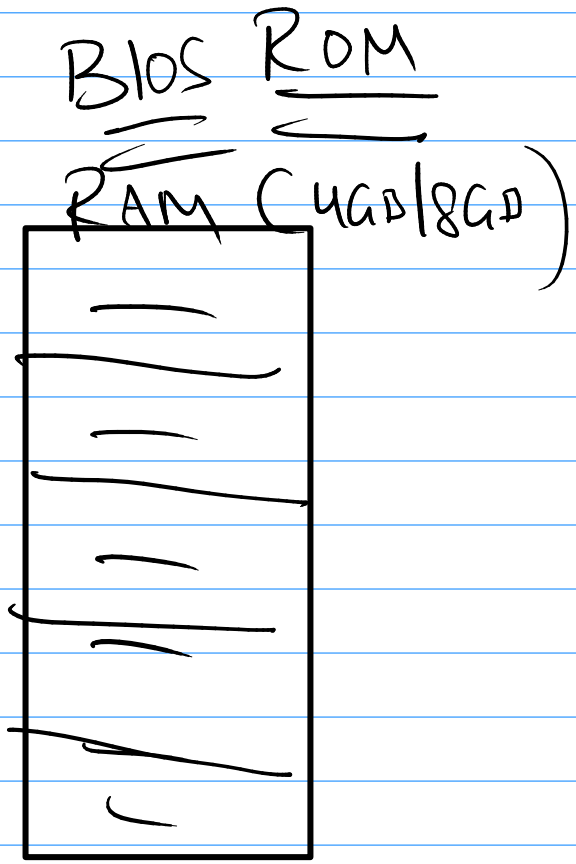
Virtual Memory :-

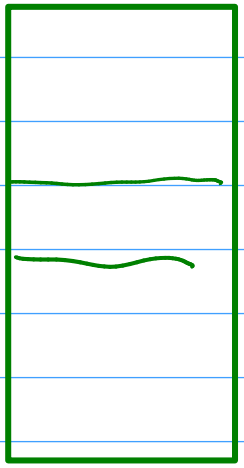
uC
without VM

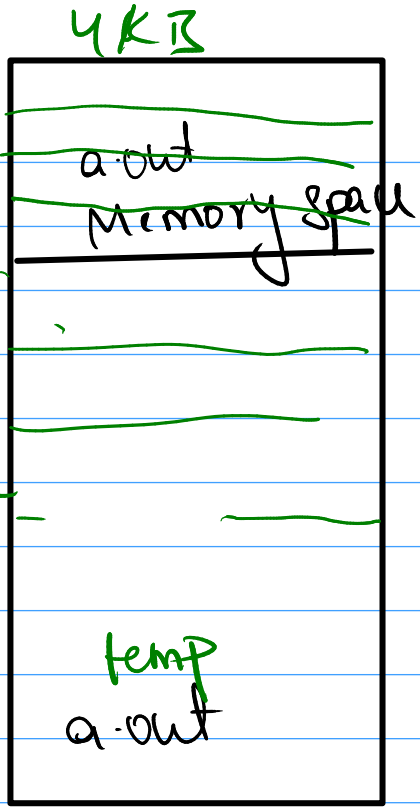


Desktop :-



$./a.out





CPU

Demands the page

page fault

trap

OS(mmu)

PCB → memory limits

VM → RAM (into the frame)

page table

Page No	frame
1	0

$./a.out

→ 10 sec

$./a.out Ready State

→ If page faults are more then exec becomes slow.

Page Replacement algo

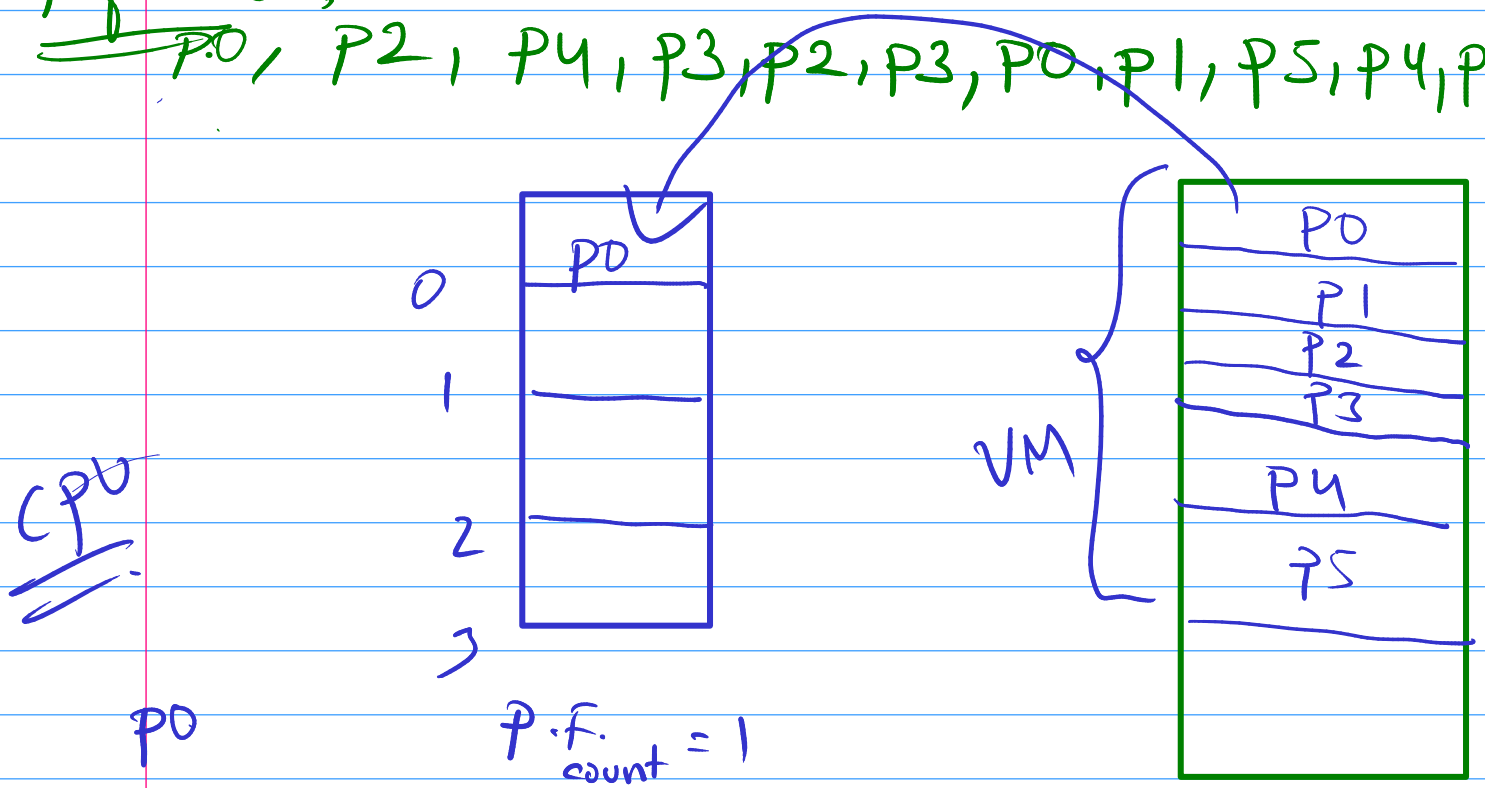
1) FIFO (First in - First out)
2) LRU (Least recently used)
3) optimal page replacement

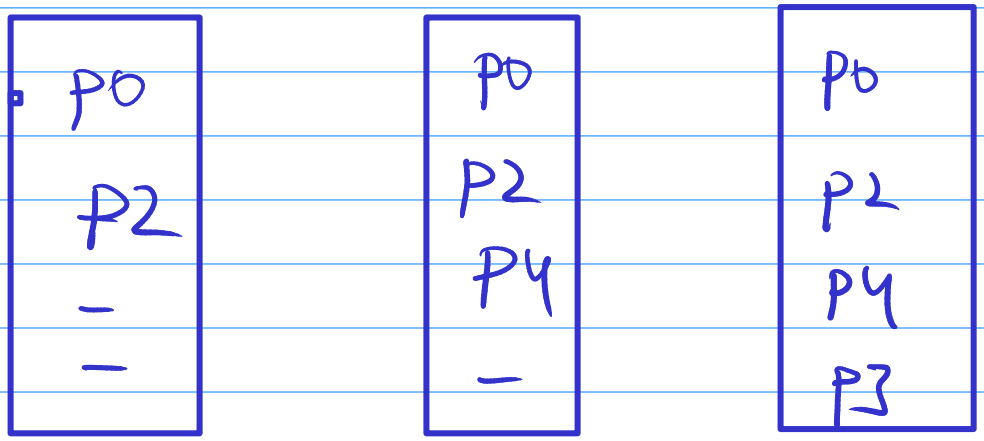
FIFO :-

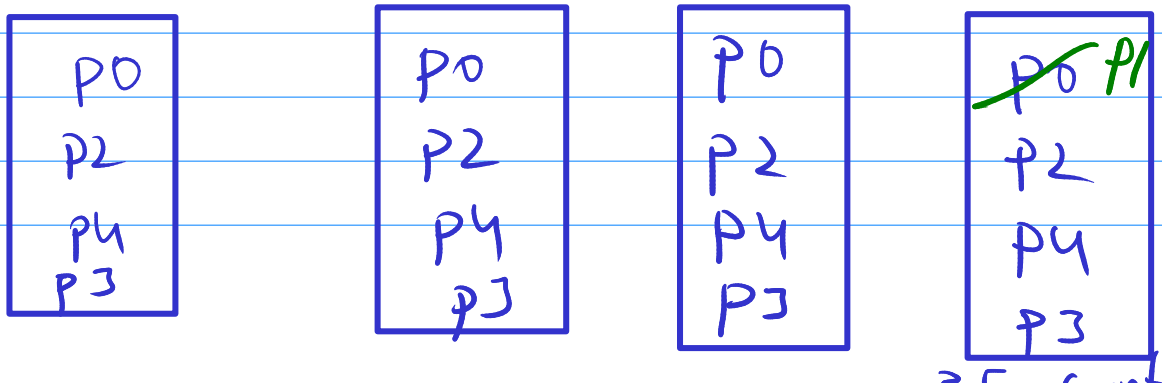
No. of frames = 4
each frame Size = 4KB
} Total RAM 16KB

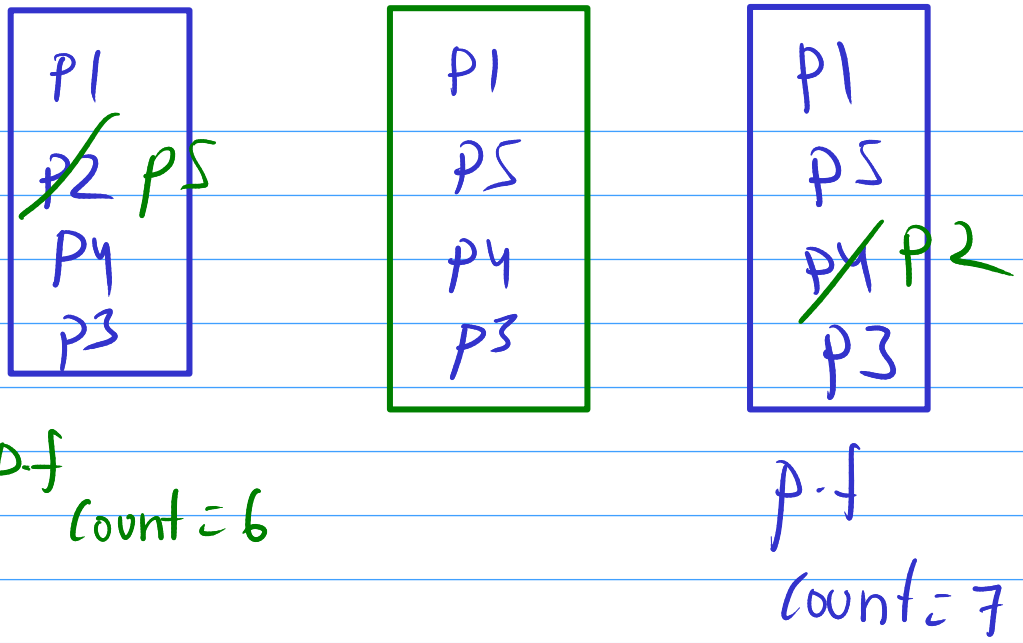
Reference :-

P0, P2, P4, P3, P2, P3, P0, P1, P5, P4, P2









P.f
Count = 6

P.f
Count = 7

Q 5 frames :-

Reference:

P0, P2, P0, P3, P4, P2, P1, P4, P5, P6, P3, P2, P0

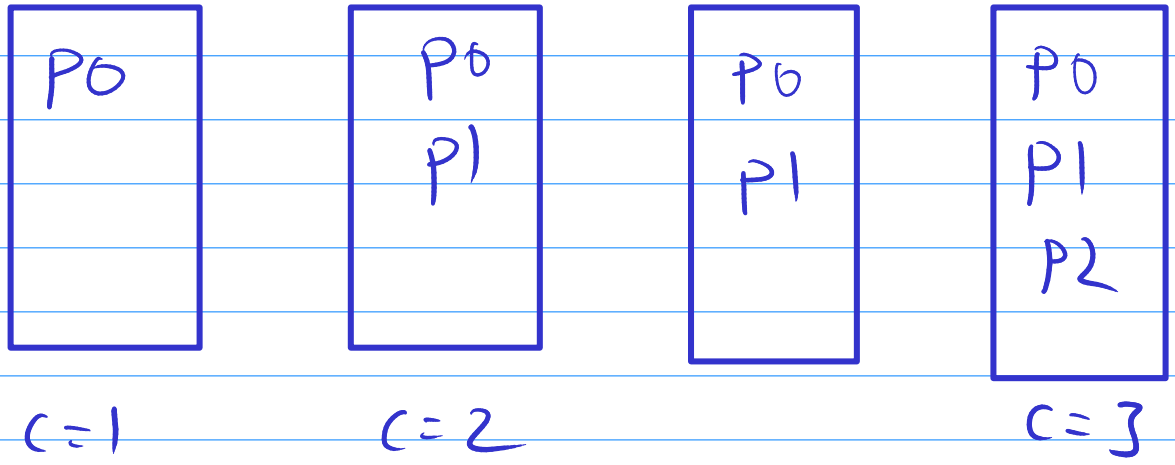
P0	P0	P0	P0	P6	P0	P6
	P2	P2	P2	P2	P2	P2
			P3	P3	P3	P3
				P4	P4	P4

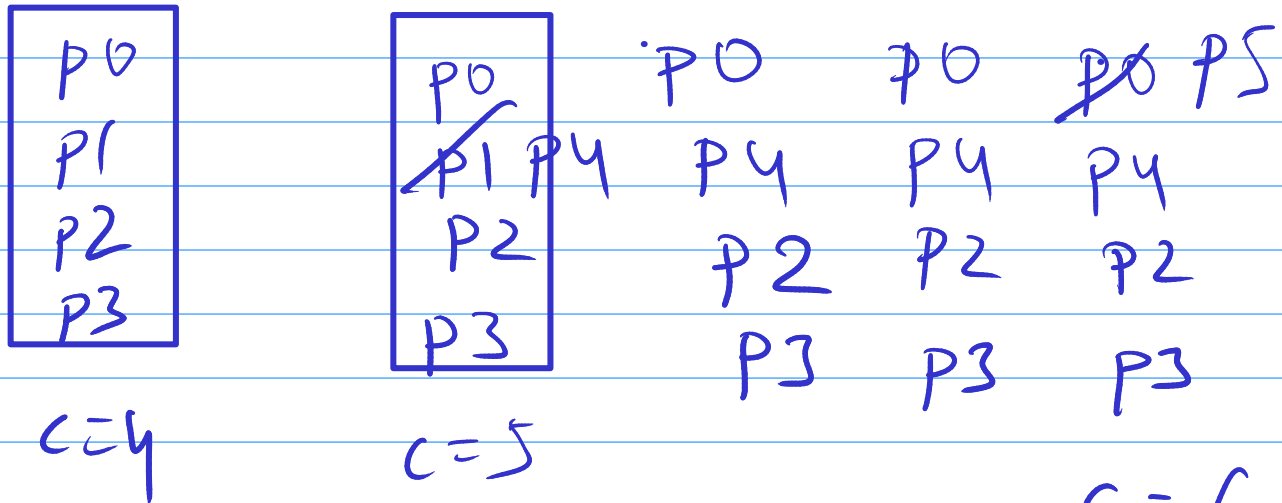
P0	P6 P5	P5	P5	P5		P1
P2	P2	P2 P6	P6	P6		P6
P3	P3	P3	P3	P3 P2		P2
P4	P4	P4	P4	P4		P4 P0
P1	P1	P1	P1	P1		P3

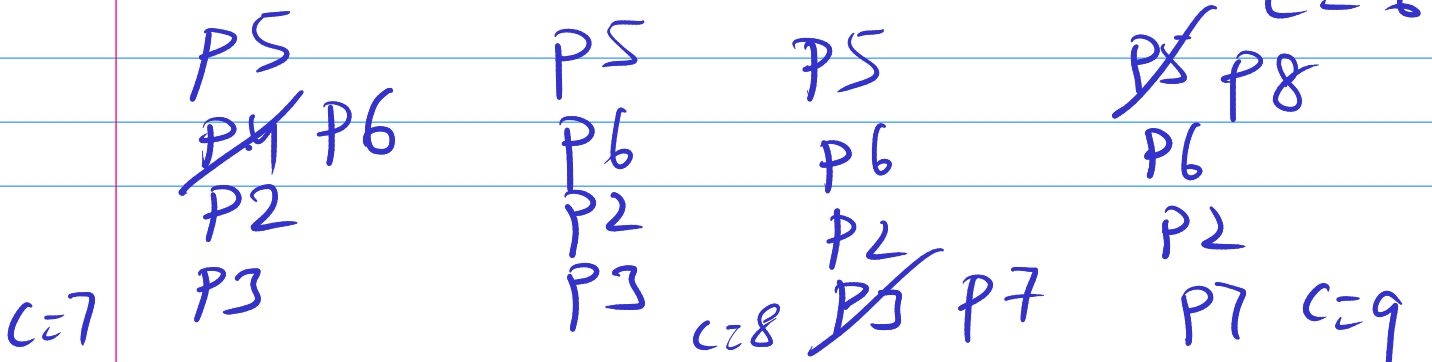
Total page faults = 9

LRU :- Total frames = 4

→ P0, P1, P0, P2, P3, P4, P3, P2, P5, P6, P2, P7, P8



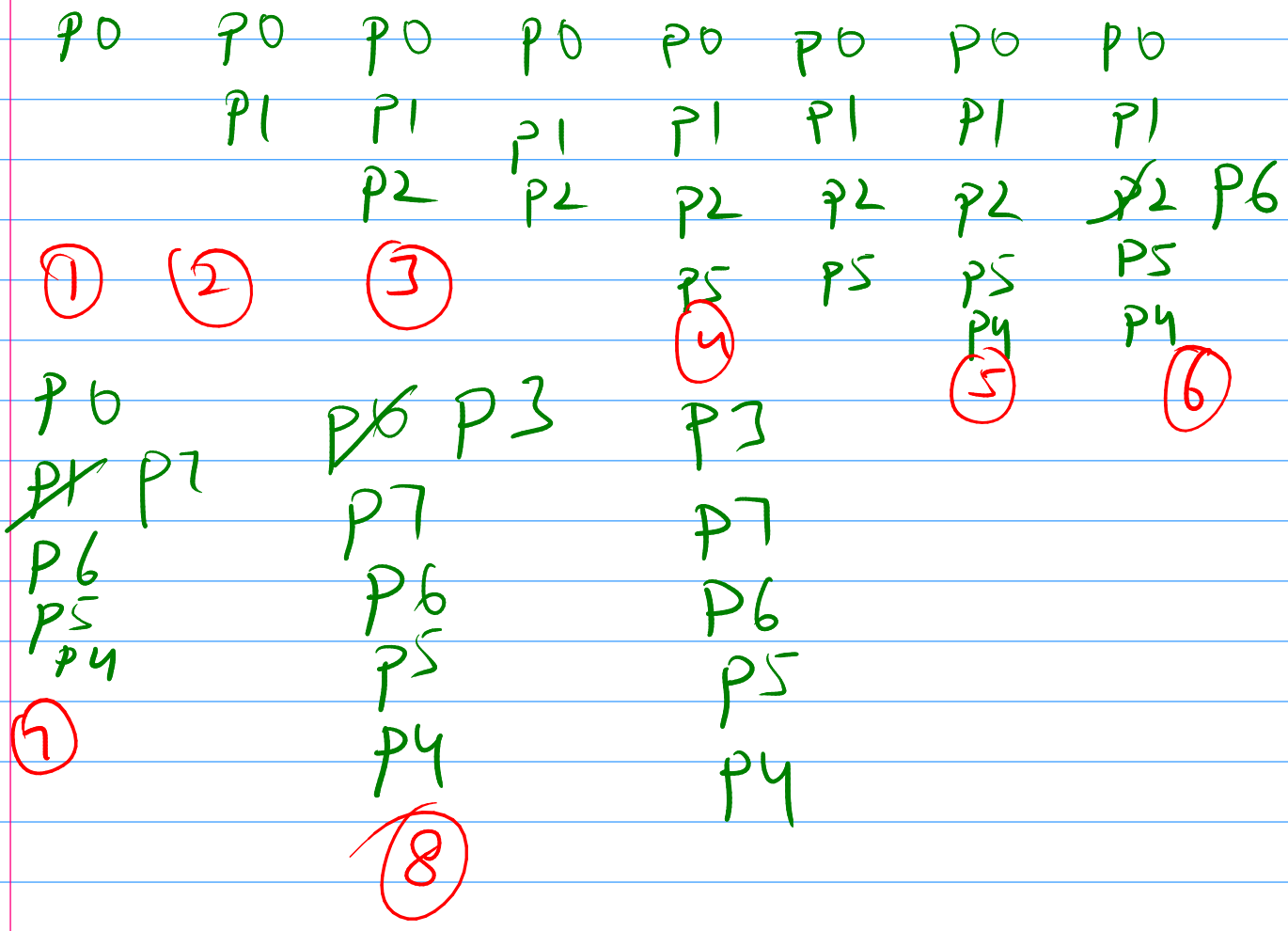




5 frames LRU

Q P0, P1, P2, P1, P5, P0, P4, P5, P6, P7, P5, P3, P7

Find No. of page faults:-



optimal page replacement algo :-

a) 4 frames

b) P0, P1, P2, P4, P5, P2, P0, P2, P6, P7, P3, P1, P0, P2

P0	P0	P0	P0	P0	P0	P0
	P1	P1	P1	P1	P1	P1
		P2	P2	P2	P2	P2
			P4	~~P4~~ P5	P5	P5

P0	P6	P0	P6
P1	P1	P1	P1
P2	P2	P2	P2
P5	~~P5~~ P6	~~P6~~ P7	~~P1~~ P3

P.f = 8